
PRINCIPAL BUNDLE GEOMETRY OF QUALIA: UNDERSTANDING THE QUALITY OF CONSCIOUSNESS FROM SYMMETRY

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ABSTRACT

Qualia, the subjective qualities of experience, pose a fundamental challenge to scientific explanation. A promising recent approach is to characterize qualia by their relational structures rather than their intrinsic nature. Assuming the structure of qualia is characterized by the geometry of neural representations, we posit that this geometry is fundamentally organized by symmetry. We propose that the principal bundle, induced by a symmetry group G acting on a neural network's state space, is the essential mathematical object that provides a complete and rigorous characterization of this geometric structure. The core mathematical principle is that, for a G -equivariant network, its state space partitions into a quotient space Q and orbits induced by the group G . This framework reveals a three-level hierarchy for characterizing qualia. First, the choice of the symmetry group G itself defines the fundamental nature of the qualia modality (e.g., vision versus audition). Second, we identify a qualia signature with a point in the quotient space Q (e.g., perceiving the signature "cat"), representing the invariant identity of a percept. Third, we identify a qualia attribute with a position on a specific orbit (e.g., the cat's location), representing its continuous variations. This framework uncovers a powerful duality: the geometry of the orbits is rigid, inheriting the structure of the group G , which explains the stable relational structure of qualia attributes. In contrast, the geometry of the quotient space is plastic, allowing the relationships between qualia signatures to be shaped by learning. This

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geometric hierarchy provides a unified language for explaining the structure of perceptual experience. Furthermore, our framework generates empirical predictions and a clear research strategy for characterizing qualia, offering a path to a deeper scientific understanding of subjective experience.

Introduction

The qualitative nature of our conscious experience, qualia, presents a profound scientific puzzle, organized across multiple levels of distinction between our experiences. In our usage, qualia are simply the qualities of subjective experience, the “what it is like” character of conscious states such as “what it is like to see a red” or “what it is like to hear a C note” [1]. At the most fundamental level, why do visual experiences feel distinctly visual and auditory experiences distinctly auditory? Within vision alone, why does perceiving the color “red” have its own unique character, distinct from “green”? How can we perceive a single, invariant object, such as a red apple, while its specific attributes, such as its location in our visual field or its orientation, continuously vary? Explaining how our brains generate such distinctive qualities of the experiences, qualia, is a central challenge of objective scientific investigation [2, 3, 4, 5, 6, 7]. Current theories of consciousness often focus on identifying the neural correlates or the functional roles of conscious states [8, 9, 10], proposing various functional mechanisms and systems-level properties of neural activity that are thought to be necessary and sufficient for conscious perception [11, 12, 13, 14, 15, 16, 17]. However, many of these theories largely do not specify the principles that determine why a particular neural activity pattern should feel like “seeing red” as opposed to “hearing a C note,” nor do they explain the underlying relational structure that organizes these experiences.

To objectively and quantitatively characterize qualia, we adopt a structural approach that characterizes qualia by their relational structures. We assume these qualia structures are determined by the geometry of their underlying neural representations. This aligns with a growing movement that seeks a characterization of qualia, grounded in the geometric structure of a system [18, 19, 20, 21, 22, 23, 24]. A key empirical motivation for focusing on the geometry of neural representations also comes from the pioneering “rewired ferret” experiments [25, 26, 27, 28]. These studies demonstrated that when the auditory cortex receives input from the eyes through surgical rewiring, it develops functional properties similar to the visual cortex, and the animals learn to perceive the resulting experiences as visual. These results suggest that the character of a qualia modality is determined not by the sensory organ, but rather by the computational structure, or the function and effective connectivity of the neural circuits processing the information. This strongly motivates a direct investigation into the geometry of neural representations.

We propose that the key to understanding neural geometry and its relation to qualia is symmetry, a principle that has proven fundamental to learning in both biological and artificial systems [29, 30, 31, 32]. The pioneering application of symmetry to perception dates to the seminal work of Hoffman, who used Lie group theory to explain visual constancies [33, 34, 35]. Building on these foundational works, our goal is to develop a more general

theory applicable across diverse sensory modalities by leveraging modern computational neuroscience and geometric deep learning. To formalize this theory, we employ the mathematical framework of a principal bundle [36, 37] (Figure 1). This structure naturally arises from any equivariant neural encoder (G), which partitions the neural state space (Y) into two components: orbits ($\mathcal{O}$), representing continuous perceptual variations generated by the group’s action, and the quotient space ($Q = Y/G$), representing dimensions of variation orthogonal to the group action. Decomposing a representation space into the orbits of a group action and a quotient space of invariants is a central concept in modern geometric theories of perception and recognition [38, 39, 40, 41, 42, 43, 44].

Our central hypothesis is that this principal bundle geometry fully characterizes the relational structure of qualia. Given this geometry, the three-level hierarchy of qualia naturally emerges (Figure 1).

- **Qualia modality:** Defined by the algebraic structure of the symmetry group G itself. The non-isomorphic nature of the groups that govern different senses (e.g., $SO(2)$ for hue versus $\mathbb{R}$ for pitch) accounts for the fundamental differences between sensory worlds.
- **Qualia signature:** Defined by a point in the bundle’s base space, the quotient space ($Q = Y/G$). This represents the G -invariant identity of a percept (e.g., the signature ‘cat’).
- **Qualia attribute:** Defined by a point on a specific orbit or fiber $\mathcal{O} \subset Y$. This represents the continuous, G -variant parameters of a percept (e.g., the specific location of the ‘cat’).

The primary contribution of this framework is the exposure of a fundamental duality: the geometry of qualia *attributes* is rigid, while the geometry of qualia *signatures* is plastic. The orbits ($\mathcal{O}$), which encode attributes, inherit their topological structure directly from the fixed, universal symmetry group (G). In stark contrast, the quotient space (Q), which encodes signatures, is a learnable manifold whose geometry is actively shaped by experience to reflect the statistical structure of the environment.

For instance, if two people acquire the same equivariance properties, the relational structure of their qualia attributes must be topologically equivalent. In contrast, there is no theoretical guarantee that the geometric relationships between their qualia signatures are similar, as this structure is shaped by each individual’s unique learning history. The following sections will introduce the mathematical concepts underlying our theory, derive the consequences of the principal bundle geometry that lead to this fundamental duality, generate empirically testable predictions, and outline a research strategy for its verification.

Symmetry groups and equivariance of neural representation

To formalize the framework presented in the introduction, we will now define the essential mathematical concepts that form its foundation. This section introduces symmetry groups,

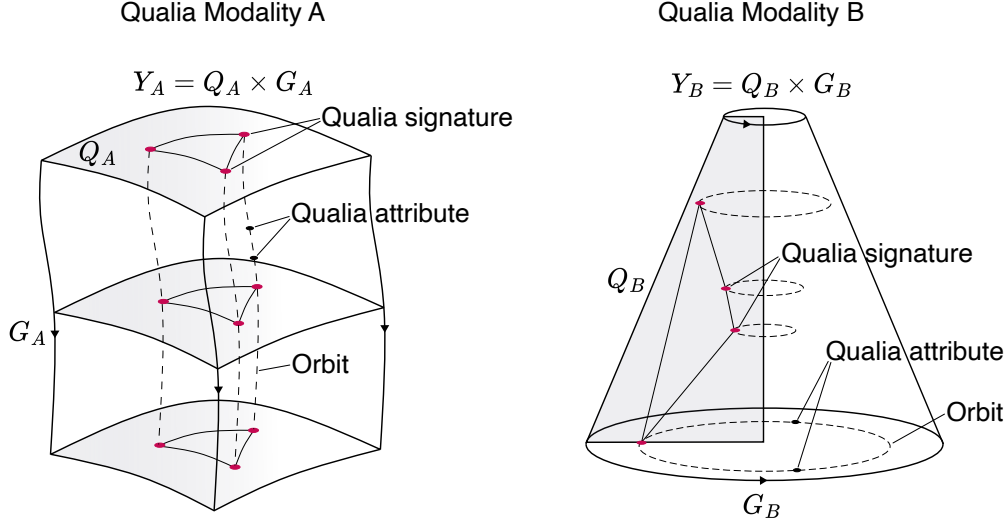


Figure 1: Hypothesis for the principal-bundle geometry of qualia. **Qualia modalities** A and B differ in their groups G_A and G_B , which reflect different invariant signal transformations. The bundle for each qualia modality is the product of the quotient space Q and the symmetry group G (arrows orthogonal to Q), which encodes transformations that leave the identity in Q invariant. The multiple shaded planes shown are cross-sections, each representing a copy of the quotient space. Thus, the bundle structure of each modality varies according to the transformations that preserve the identity in Q . **Qualia signatures** are defined by red points in Q . These points represent the G -invariant identities of different percepts (e.g., the signatures “cat” and “dog”). **Qualia attributes** are defined by points on a specific orbit (a dashed line). These represent the continuous, G -variant parameters of a percept (e.g., the specific locations of the “cat”).

group actions, and equivariance of neural representations that is central to our framework. The next section demonstrates how these concepts lead to the geometric structures of orbits and quotient spaces in neural representation space. Together, these elements provide a comprehensive description of our hypothesis.

Symmetry groups

At the heart of our theory lies the concept of a symmetry group. Intuitively, **symmetries describe transformations that leave a key aspect of an object or system unchanged. A group is the mathematical formalization of this concept.** It is a set of transformations, denoted G , that includes an identity transformation (which does nothing), where any two transformations can be combined to form a third transformation, and every transformation has an inverse. For instance, the set of all possible translations of an image on a 2D plane forms a group because any two translations can be combined, and every translation has an inverse. This group can be denoted as $G = \mathbb{R}^2$.

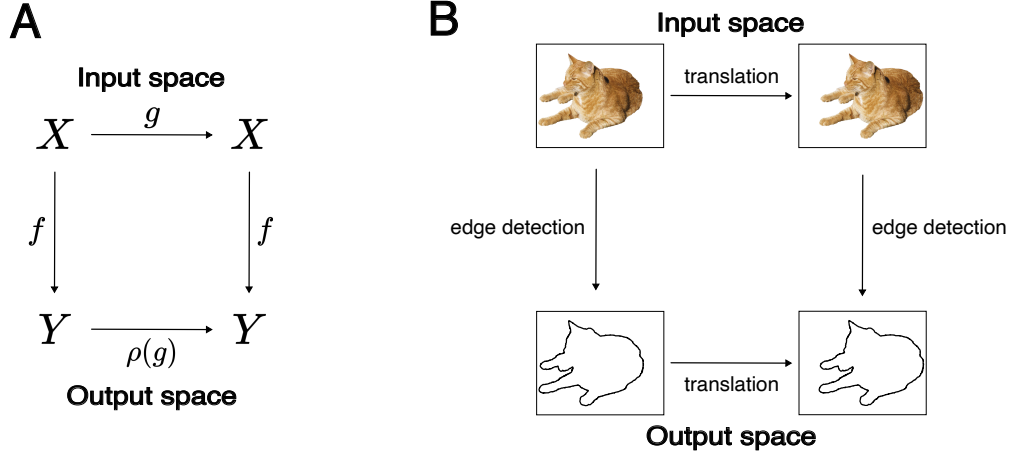


Figure 2: Definition and toy example of equivariance. (A) Commutative diagram for (G, ρ) -equivariance. The input transformation of a group element g and the corresponding output transformation $\rho(g)$ run horizontally, while the function $f : X \rightarrow Y$ (neural networks) acts vertically. The diagram commutes if and only if $f(g \cdot x) = \rho(g) f(x)$ for all $g \in G, x \in X$. (B) A toy example illustrating equivariance in image processing. Applying a translation to the right on the input image, i.e., a lying cat, followed by edge detection, yields the same result as applying edge detection first, followed by a translation to the right on the output. This demonstrates that the edge detector f is equivariant with respect to the translation group.

Group actions

A group becomes meaningful when it acts on a space, meaning each transformation in the group maps points in the space to other points within the same space. This is how abstract symmetries manifest as concrete changes. In our framework, we consider two such actions:

1. **Action on the stimulus space (X):** This is the world of sensory inputs. For a stimulus $x \in X$ (e.g., an image of a cat) and a group element $g \in G$ (e.g., a specific translation), the action $g \cdot x$ produces the transformed stimulus (the translated image).
2. **Action on the neural state space (Y):** This is the world of neural activity. For a neural activity pattern $y \in Y$, a group element g also induces a transformation, denoted by $\rho(g)$. The transformed neural state is thus $\rho(g)y$. The function ρ is called a representation of the group. It translates the abstract group elements into concrete operations within the neural state space.

Equivariance: linking stimulus and neural symmetries

The crucial link between the stimulus and the neural world is a property called equivariance. A neural encoder, $f : X \rightarrow Y$, is said to be G -equivariant if the two paths, visualized in Figure 2, lead to the same result:

- **Transform then encode:** Applying a transformation $g \in G$ to the input x and then encoding the result corresponds to computing $f(g \cdot x)$.
- **Encode then transform:** Encoding the input x first and then applying the corresponding transformation to the neural representation corresponds to computing $\rho(g) \cdot f(x)$.

This principle of consistency is captured by the following equation:

$$f(g \cdot x) = \rho(g) \cdot f(x) \quad \text{for all } g \in G, x \in X \quad (1)$$

Intuitively, the property of equivariance forces the neural representation space Y to be organized by the symmetries of the group G . As the next section will elaborate, this imposed geometric organization gives rise to a characteristic structure known as a principal bundle, which provides the mathematical foundation for our theory of qualia.

Note that invariance is a special case of equivariance in which the representation is the identity, i.e., $f(g \cdot x) = f(x)$. As will be explained later, characteristic principal bundle geometry does not appear in the case of invariance. Thus, this manuscript is mostly concerned with cases of equivariance where the group representation ρ is not the identity (more precisely, ρ acts freely on Y).

Equivariance in neuroscience and AI

The principles of equivariance and invariance are cornerstones of both biological and artificial neural computation. The early visual cortex (V1) contains retinotopic maps that represent stimuli in an explicitly equivariant manner: a translation of a stimulus on the retina causes a corresponding translation of the activity pattern on the cortical map [45, 46, 47].

[48]. As information progresses hierarchically to the inferior temporal (IT) cortex, neurons develop representations of object identity that are remarkably invariant to changes in position, scale, and viewpoint, as revealed by both single-unit recordings and human fMRI studies [49, 50, 51, 52, 53, 54]. This progression towards invariance, however, does not mean that attribute information is discarded. While some ventral stream neurons achieve high selectivity for object identity, other neural populations continue to encode properties like position and color. Crucially, the representation of these attributes varies systematically with the specific transformations applied, thus preserving the information needed for a complete perceptual experience [55, 56, 57, 58].

This strategy of progressing from equivariant to invariant representations is considered a general feature of sensory processing.

[59]. Similarly, the somatosensory cortex contains topographic body maps that are equivariant to the location of a tactile stimulus on the skin [60]. In each modality,

Inspired by these biological principles, artificial intelligence has achieved tremendous success by incorporating symmetry into network architectures. The most prominent example is the convolutional neural network (CNN), whose shared-weight structure explicitly enforces translational equivariance, a powerful inductive bias that dramatically improves learning efficiency and generalization [61, 62]. This core idea has been generalized in the field of Geometric Deep Learning, which develops models that respect other symmetries, such as rotation and reflection, by incorporating the mathematics of group theory [32, 40, 63, 64, 65]. This convergence has created a powerful synergy between the fields. Equivariant deep learning models are no longer just inspired by the brain; they are now used as state-of-the-art in silico models to understand it (cortical visual processing: [66, 67, 68]; subcortical visual processing: [69, 70]; cortical auditory processing [71, 72]). Recent work has shown that the ability of different AI models to explain the neural activity patterns of primates and the voxel activation patterns of humans in their ventral stream is strongly correlated with how well those models learn invariant object representations [73, 74]. This tight loop, where neuroscience inspires AI, and AI, in turn, provides testable models of the brain, underscores that equivariance is a canonical principle of both biological and artificial intelligence.

The central hypothesis: principal bundle geometry of qualia

When a neural encoder is equivariant with respect to a symmetry group G , this symmetry naturally organizes or “decomposes” the neural state space Y into a structure known as a principal bundle (see Supplementary Information for a formal definition). This structure consists of two fundamental components: orbits and a quotient space (Figure 1).

Our central hypothesis is that this principal-bundle geometry mathematically characterizes qualia organized across a three-level hierarchy.

Symmetry groups: The identity of qualia modalities. At the most fundamental level, our hypothesis posits that the algebraic structure of the symmetry group G itself determines the nature of a qualia modality. Assume that neural encoders for different sensory modalities, such as vision and audition, acquire equivariance with respect to different symmetry groups, G_A and G_B . For generality, each of these can be a composition of multiple symmetry groups; the core assumption is that the overall structure of G_A is non-isomorphic to G_B . Because orbits inherit their topology directly from their generating group (as formalized later in Proposition 1), these different groups will induce orbits with fundamentally different topological structures (Figure 1). Conversely, all orbits generated by the same group are topologically equivalent to each other (Proposition 2). We posit that this difference in orbit topology, rooted in the non-isomorphic nature of the underlying symmetry groups, corresponds to the fundamental phenomenological difference between sensory worlds.

Orbits: The geometry of qualia attributes. For any given neural state y , the set of all states reachable by applying transformations from G forms the orbit of y . Mathematically, this is written as

$$\mathcal{O}(y) = \{\rho(g) \cdot y \mid g \in G\} \quad (2)$$

We propose that an orbit $\mathcal{O}(y)$ is a submanifold within the neural space that can be thought of as a pathway containing all possible changes in the attributes of an experience corresponding to y (Figure 1). Thus, a specific point on an orbit represents a particular qualia attribute, while the entire orbit represents the full range of possible attributes for a given signature. For instance, one orbit could consist of all the neural states corresponding to a specific cat, seen at every possible position in the visual field. Moving along an orbit changes the sensory parameters of the experience, i.e., its attributes, such as location or orientation, but not the core identity of what is being perceived.

Quotient Space: The geometry of qualia signatures. If we treat each entire orbit as a single abstract point, the collection of all these points forms a new space called the quotient space, written as $Q = Y/G$. This is the space that remains after we “quotient out,” or disregard, the variations that occur within each orbit. There is a natural projection map, $\pi : Y \rightarrow Q$, that takes any point y in the full neural space and identifies which orbit it belongs to.

We propose that the quotient space Q is the space of qualia signatures (Figure 1). Each point in Q represents a distinct, G -invariant identity of an experience, such as “cat” and “dog,” without regard for their specific, variable attributes.

Phenomenology of qualia signatures and attributes in different modalities

The principal-bundle framework provides a unified explanation for the structure of diverse sensory experiences. Here, we illustrate how the geometric decomposition maps onto the phenomenology of different qualia modalities, such as vision and audition. These examples demonstrate that the specific algebraic structure of the relevant symmetry group G defines the modality, while the corresponding orbits and quotient spaces define its attributes and signatures. This theoretical approach finds strong support in a variety of findings from perceptual neuroscience, which reveal how neural systems build representations that are tolerant to some transformations while remaining sensitive to others [31, 75].

Visual objects under translation This example considers how we perceive an object’s signature as distinct from its spatial attributes (Figure 3A). The governing symmetry group for translation is $G = \mathbb{R}^2$. This decomposition has a direct parallel in the functional architecture of the primate visual system, famously described by the “two-streams hypothesis,” which posits a ventral stream for object identification (“what”) and a dorsal stream for spatial processing (“where”) [76, 77]. This model is supported by neurophysiological evidence showing that the inferior temporal (IT) cortex builds position-tolerant object representations [49, 52] and by psychological phenomena demonstrating that location serves as a robust perceptual cue for object identity [78].

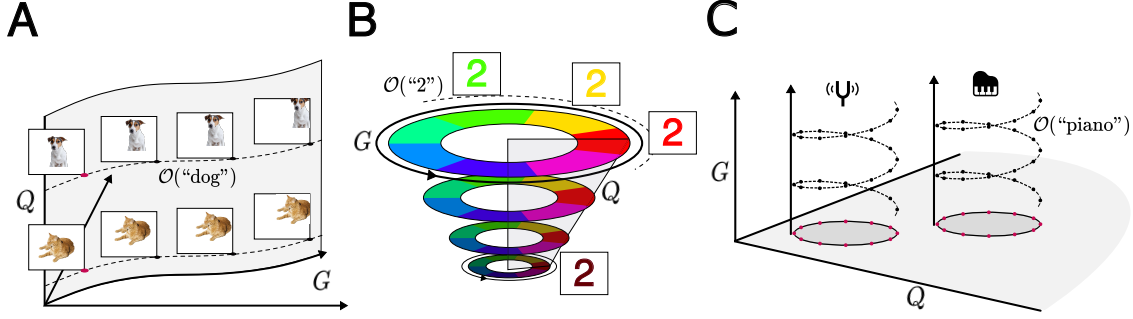


Figure 3: Mapping principal bundle structures onto the phenomenological structures of experiences in different modalities. (A) The translation group in visual objects. Qualia signatures in Q correspond to object identities (“dog” and “cat”). Qualia attributes correspond to changes in a curved line orbit (a dashed line, e.g., $\mathcal{O}(\text{“dog”})$) under the action of G , representing spatial instantiations of the same signature (“the dog at the center”). (B) The hue-rotation group $G = \text{SO}(2)$ in colors. Qualia signatures in Q correspond to changes in color saturation and lightness. Qualia attributes correspond to changes in a circular orbit (rainbow ring along with a dashed line, e.g., $\mathcal{O}(\text{“2”})$), representing different hues of the same color at a fixed saturation and lightness. (C) The pitch shift group $G = \mathbb{R}$ in sounds. Qualia signatures in Q correspond to timbre identities (“pure tone” on the left and “piano note” on the right). These signatures form pitch-class circles for specific timbres (red markers). Qualia attributes correspond to changes in a helical orbit, e.g., $\mathcal{O}(\text{“piano”})$, representing different pitches of the same timbre.

- **Qualia signatures:** The quotient space contains points corresponding to object identities, which are invariant under position translation. Points in this space represent qualia signatures such as “dog” and “cat” (Figure 3A).
- **Qualia attributes:** For a given signature (an object identity), the orbit generated by the $\mathbb{R}^2$ action is a 2D space. Points in an orbit represent qualia attributes, i.e., the specific locations of the same object (e.g., “the dog at the center” versus “the dog on the right”) (Figure 3A).
- **Phenomenology:**
 - **Change in signature:** If the object itself morphs (e.g., from a cat into a dog), the neural state jumps from the “cat” orbit to the “dog” orbit. The perceived qualia signature changes accordingly.
 - **Change in attribute:** As an object moves across the visual field, the neural state glides smoothly along its orbit. The experience of location attributes changes continuously while the qualia signature (e.g., “dog”) remains the same.

Colors under hue rotation This example addresses the circular nature of color perception (Figure 3B). In a standard color model such as HSL (Hue, Saturation, Lightness), changes in hue correspond to rotations around a central axis. This symmetry is modeled with the symmetry group of 2D rotations, $G = \text{SO}(2)$. This geometric model is supported by both

neurophysiology and psychophysics. Visual area V4 exhibits “hue maps,” where neurons tuned to different hues are arranged in a circular pattern. These neurons encode hue as a continuous circular dimension. This systematic representation is inherited by IT neurons [79, 80], aligning with perceptual maps of color similarity derived from human judgment [81].

- **Qualia signatures:** The quotient space contains points corresponding to color properties, namely saturation and lightness, which are invariant under hue rotation. Points in this space represent qualia signatures such as “pure, bright colors” and “dark, pastel colors”.
- **Qualia attributes:** For a given signature (a fixed saturation and lightness), the orbit generated by the $SO(2)$ action is a circle, topologically an S^1 . Points on a circle orbit represent qualia attributes, i.e., specific hues (e.g., “scarlet” and “indigo”).
- **Phenomenology:**
 -  (Figure 3B).
 - **Change in attribute:** As color hues change, the neural state continuously changes along its S^1 orbit. The experience moves through the rainbow: red $\rightarrow$ orange $\rightarrow$ yellow, etc., while the qualia signature (e.g., “bright, pure color”) remains fixed (Figure 3B).

Sounds under pitch shift This example considers how the auditory system identifies a stable sound signature (timbre) across variations in its pitch attribute (Figure 3C). The action of pitch change is modeled by the symmetry group $G = \mathbb{R}$. The core of this model is that the brain possesses a representation of timbre that is tolerant to the G -action of pitch transposition. This is consistent with our fundamental ability to recognize a musical instrument, such as a violin or a piano, regardless of whether it is playing a high or low note [82]. While pitch and timbre can interact perceptually [83, 84], neuroimaging and behavioral studies support the brain’s capacity to disentangle them [85, 86]. These findings support a geometric model where timbre acts as the qualia signature, forming a stable point in a quotient space, while pitch is the variable qualia attribute, corresponding to a position along an orbit.

- **Qualia signature:** The quotient space contains points corresponding to timbral identities, such as “steady pure tone” and “piano note”.
- **Qualia attribute:** For a given signature (a timbre), an orbit generated by the $\mathbb{R}$ action is represented as a line or a helix. Points on a helix orbit represent qualia attributes, i.e., specific pitches.

- **Phenomenology:**

- **Change in signature:** If harmonics are added to a pure tone, the neural state moves from the “pure tone” orbit to the “complex timbre” orbit. The perceived qualia signature changes accordingly (Figure 3C).
- **Change in attribute:** As the pitch of a timbre shifts, the neural state moves continuously along the helix orbit. The experience of pitch attribute continuously changes while the qualia signature (e.g., “piano note”) remains the same (Figure 3C).

The duality of qualia geometry: Rigid attributes and plastic signatures

In this section, we develop the geometric implications of our framework. Our central claim is that the principal bundle geometry induced by the symmetry group imposes a powerful duality on the structure of qualia: the geometry of qualia attributes is topologically rigid and universal, while the geometry of qualia signatures is plastic and learnable. We first formalize the conditions for this structure to arise. Then, we mathematically prove the rigidity of the orbits, representing qualia attributes, and the plasticity of the quotient space, representing qualia signatures.

We will now establish this duality through a series of propositions. Propositions 1-4 will establish the rigidity of the orbits, showing that they are topologically identical across subjects and can be metrically rigid under stronger conditions. In contrast, Propositions 5-7 will address the plasticity of the quotient space, demonstrating that its geometry can be learned and vary across subjects, while remaining coherent from different viewpoints within a subject.

In the following, we assume that the symmetry group G is a Lie group [87] acting smoothly, properly, and freely on an input space X . The neural representation space Y is a d -dimensional Euclidean space, $Y \simeq \mathbb{R}^d$. The group acts freely on Y via a continuous linear representation $\rho: G \rightarrow GL(d)$. The neural encoder $f: X \rightarrow Y$ is assumed to be exactly equivariant, satisfying $f(g \cdot x) = \rho(g)f(x)$.

Rigidity of orbits: The universal structure of qualia attributes

The equivariance of the encoder f has profound consequences for the geometry of the orbits. It ensures that their fundamental topological structure is not learned or arbitrary but is rigidly inherited from the symmetry group G itself. This rigidity can be formalized in the following propositions (see Supplementary Information for the proofs).

Proposition 1 (Orbit-group diffeomorphism; Figure 4A). Let $f: X \rightarrow Y$ be a smooth, proper, G -equivariant map between smooth G -spaces X and Y . Assume the action on X is free and proper. Then for any $x_0 \in X$, if the action of G on Y is free at $y_0 = f(x_0)$, the orbit $\mathcal{O}_Y(y_0)$ is a properly embedded smooth submanifold of Y that is diffeomorphic to G .

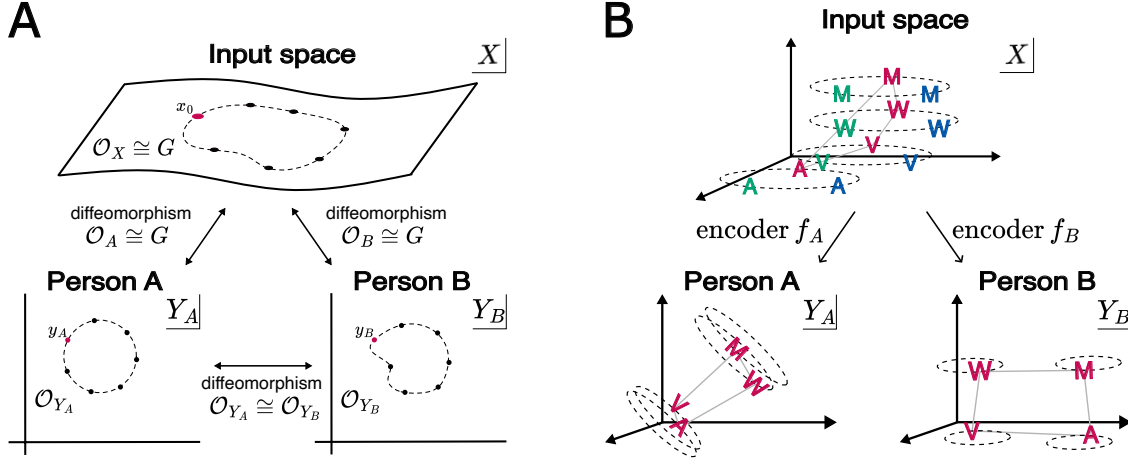


Figure 4: Rigidity of qualia attributes and plasticity of qualia signatures. (A) Rigidity of qualia attributes. Any orbit formed by the group action G in the neural representation space Y is diffeomorphic to G (Proposition 1). Orbits in person A and person B, $\mathcal{O}_{Y_A}(y_A)$ and $\mathcal{O}_{Y_B}(y_B)$, are diffeomorphic (Proposition 2). This implies that the structure of qualia attributes, as represented by the structure of orbits, is topologically rigid and universal across subjects. (B) Plasticity of qualia signatures. The symbols “A,” “M,” “V,” and “W” form orbits under the group action G in the input space X , which corresponds to color changes. A copy of the quotient space represents the relational structures of the red color symbols “A,” “M,” “V,” and “W,” for example. The relational structures formed by the red color symbols can differ between the input space and the neural representation space (Proposition 5). Thus, in general, persons A and B have different relational structures in the quotient space (Proposition 6). This implies that the structure of qualia signatures, as represented by the relational structure in the quotient space, is plastic and subject-specific.

This proposition establishes a strong topological rigidity. The general result of topological rigidity leads to a powerful conclusion about comparing qualia across different individuals or networks.

Proposition 2 (Inter-subject orbit diffeomorphism; Figure 4A). Let Person A have an equivariant encoder $f_A : X \rightarrow Y_A$ (with representation ρ_A) and Person B have an equivariant encoder $f_B : X \rightarrow Y_B$ (with representation ρ_B), both with respect to the same Lie group G . Let $x_0 \in X$ be a point in the stimulus space, and let $y_A = f_A(x_0) \in Y_A$ and $y_B = f_B(x_0) \in Y_B$ be the corresponding neural representations. If the G -actions on Y_A and Y_B are free at y_A and y_B respectively, then the orbit $\mathcal{O}_{Y_A}(y_A)$ is diffeomorphic to the orbit $\mathcal{O}_{Y_B}(y_B)$.

A stricter, metric rigidity is achieved under the additional condition of orthogonality. When the representation is orthogonal, the orbit inherits a canonical metric geometry from the group G itself. This means that the geometry of the group space is homogeneous, that is, the distance between any two transformations g_1 and g_2 depends only on the relative transformation $g_2^{-1}g_1$ needed to get from one to the other, not on their absolute position within the group. This provides a rigid, canonical “ruler” for measuring distances on the orbit. The following proposition makes this precise.

Proposition 3 (Orbit-group isometry). Suppose the neural representation space Y is endowed with a Riemannian metric g_Y (for example, the Euclidean metric). If the representation $\rho : G \rightarrow O(d)$ is an orthogonal representation, then for any $y_0 \in Y$ in the image of f , the orbit $\mathcal{O}_Y(y_0)$, endowed with the induced Riemannian metric, is isometric to the Lie group G , endowed with the pullback Riemannian metric $\phi_Y^* g_Y$. Furthermore, this pullback metric $\phi_Y^* g_Y$ on G is a left-invariant Riemannian metric.

By the assumption of orthogonal representation, there is a much stronger similarity between the orbits across different individuals than a mere diffeomorphism. The orthogonality condition ensures that each orbit inherits a rigid, homogeneous geometry from the group itself, where the geometry at any point is a simple translation of the geometry at the identity. Consequently, the diffeomorphism Ψ connecting two such orbits cannot be an arbitrary local warping. Instead, the distortion required to map one orbit onto the other must be globally uniform, dictated by a single linear transformation on the underlying Lie algebra. In the special case where this transformation is a uniform scaling, the orbits are related by a homothety. If this scaling factor is one, the orbits are metrically identical (isometry). This leads to the following proposition.

Proposition 4 (Uniform metric distortion of inter-subject orbit maps). Let $\mathcal{O}_{Y_A}(y_A)$ and $\mathcal{O}_{Y_B}(y_B)$ be two orbits generated by orthogonal representations. The entire geometric relationship between them is determined by a single, constant, self-adjoint linear operator $L : \mathfrak{g} \rightarrow \mathfrak{g}$ on the Lie algebra $\mathfrak{g}$, which relates the two pullback metrics at the identity: $\langle X, Z \rangle_B = \langle X, L(Z) \rangle_A$. In the special case where this operator is a scalar multiple of the identity, $L = c \cdot \text{Id}$, the map Ψ is a homothety.

Consequence for qualia attributes. These propositions provide a formal basis for the rigidity of qualia attributes. The fundamental topological structure of how attributes relate to one another (e.g., that hues are arranged in a circle) is a universal consequence of the underlying symmetry group G , as guaranteed by Proposition 1. This topological equivalence is universal across individuals; even if two subjects learn very different neural representations ($\rho_A \neq \rho_B$), Proposition 2 implies that the structure of their corresponding qualia attributes must be diffeomorphic. This constrains individual differences in experience to be, at most, smooth warpings of a universal geometric object.

A much stricter, metric rigidity is addressed in the subsequent propositions. If a neural representation is orthogonal, the orbit’s intrinsic geometry becomes a rigid, homogeneous copy of the group G itself, endowed with a canonical left-invariant metric (Proposition 3). Because both subjects’ orbits are then metrically rigid copies of the same group, the mapping Ψ that connects their qualia spaces cannot be an arbitrary warping. Instead, as described in Proposition 4, the distortion must be globally uniform, dictated by a single, constant linear operator on the underlying Lie algebra. In the special case where this operator is a simple scaling, the map is a homothety, meaning the structure of the two qualia attributes has the exact same shape.

Plasticity of quotient space: The learnable structure of qualia signatures

While the orbits that define qualia attributes have a rigid topological structure, the geometry between orbits, which defines the relationships between qualia signatures, is plastic. This plasticity allows a neural system to flexibly organize its conceptual space based on learning and task demands.

To analyze the geometry of the quotient space Q_Y , it is useful to have a concrete representation of this space within the larger neural manifold Y . Such a representation is given by a section, which is a submanifold $S \subset Y$ that intersects each orbit exactly once. For the sake of a clear geometric analysis, we will assume that a smooth global section exists. This assumption simplifies the picture by ensuring that the neural representation space Y is not globally twisted, a condition mathematically equivalent to the principal bundle being trivial, i.e., globally diffeomorphic to a product space $Y \cong Q_Y \times G$ (see, e.g. [88]). The subsequent propositions explore the geometry of this section S .

Proposition 5 (Quotient plasticity; Figure 4B). Let S be a smooth global section of Y corresponding to a G -equivariant encoder f . The intrinsic geometry of S is not fixed by the G -equivariance constraint alone. By composing the encoder f with a valid G -equivariant linear map, the geometry of the resulting section $\tilde{S}$ can be arbitrarily altered.

This plasticity explains why different individuals can learn and refine their understanding of the world in different ways, depending on task demands or experiences. This leads immediately to the next proposition, which states that the quotient space structure is person-specific:

Proposition 6 (Inter-subject quotient plasticity; Figure 4B). Let Person A and Person B have encoders f_A and f_B that are both equivariant with respect to the same group G . Let S_A and S_B be any corresponding sections of Y_A and Y_B , respectively. There is no guaranteed geometric relationship (e.g., isometry, similarity, or even diffeomorphism) between the submanifolds S_A and S_B with their inherited Riemannian metrics.

Finally, although the geometry of the quotient space is learnable, we remark that it must be coherent across different viewpoints for a single individual. This ensures that the structure of the space of signatures remains consistent, regardless of the specific attributes (e.g., location, orientation) through which they are viewed. This intra-subject coherence is formalized by the following proposition.

Proposition 7 (Intra-subject quotient diffeomorphism). Let $s_1 : Q_Y \rightarrow Y$ be a smooth global section, and let $S_1 = s_1(Q_Y)$ be its image. For any group element $g \in G$, let S_2 be the image of the section $s_2 = \rho(g) \circ s_1$. The map $\Psi_g : S_1 \rightarrow S_2$ defined by restricting the group action, $\Psi_g(y) = \rho(g)y$, is a diffeomorphism.

Consequence for qualia signatures. These propositions provide a formal basis for the plasticity of qualia signatures. Proposition 5 shows why different individuals can refine their internal structure of qualia signatures in different ways. However, note that Proposition 7 ensures that this learned map is internally coherent and stable across viewpoints within the same individual. It shows that the geometric structure of the quotient space is consistent

regardless of the specific qualia attributes (e.g., location or orientation) through which the signatures are viewed. Finally, Proposition 6 immediately follows from Proposition 5, formalizing the idea that this coherent, learned structure is not universal but idiosyncratic. Two people can learn to structure qualia signatures in radically different ways, depending on their experiences, goals, and even random variations in their neural wiring, all while sharing a universal, topologically rigid structure for the attributes within each of those signatures. This captures the duality of a shared physical world and a private, learned understanding of it (Figure 4A and B).

Comparing qualia structures across individuals

A central research topic in the study of qualia is investigating whether two individuals can have equivalent or similar subjective experiences, and, if not, how and to what extent they differ [24]. Regarding this question, our framework provides the following hypothesis based on the key property of the principal bundle: the structure of qualia attributes (orbits) is rigid and universal, whereas the structure of qualia signatures (quotient space) is plastic and person-specific. More specifically, if two people have equivariant encoders with respect to the same symmetry group G , our framework predicts that the structure of qualia attributes must be diffeomorphic, whereas there is no guaranteed relationship for the plastic quotient spaces.

This distinct duality motivates us to analyze the inter-subject similarity of qualia structures separately in terms of the geometry of qualia attributes (orbits) and qualia signatures (quotient space), which will be detailed below. To our knowledge, no such studies have been conducted to examine the inter-subject similarity of both terms separately. This novel approach has the potential to provide deeper insight into the nature of qualia structures and individual differences.

Before proceeding, it is crucial to clarify the different levels of equivalence that can exist between these geometric structures. In decreasing order of similarity, we can distinguish between: isometry, which preserves all metric properties like distances and angles; diffeomorphism, which preserves the smooth structure but may distort metric properties; and homeomorphism, which preserves only the fundamental topological properties, such as connectedness and the number of holes. Below, we propose several different analyses suitable for these different levels of similarity.

A two-stage protocol for empirical evaluation

Based on our propositions and empirical predictions, we present a two-stage protocol for evaluating the similarity of qualia structures across subjects. The first stage aims to verify the theoretically predicted universal structure of qualia attributes in orbit space. The second stage is exploratory and seeks to quantify the degree of similarity in the structure of qualia signatures in quotient space.

Stage 1: Verifying the structure of qualia attributes

The goal of this stage is to verify the core prediction that the structure of qualia attributes, represented by orbits, is diffeomorphic across individuals (Proposition 2). A diffeomorphism is a strong form of “smooth equivalence,” which preserves not only the topology of the orbits but also their differentiable structure. This is a stricter condition than mere topological equivalence (homeomorphism). If the representation is orthogonal, even isometry, the strongest form of geometric equivalence, could potentially be observed as a special condition (Proposition 4).

We propose a multi-faceted approach that first tests for topological equivalence as a necessary baseline, and then uses methods sensitive to the richer geometric structure preserved by a diffeomorphism.

- **Evaluating topological equivalence.** As a necessary first step, we verify that the orbits are topologically equivalent (homeomorphic) using tools from Topological Data Analysis (TDA), particularly persistent homology [89]. This method computes a unique topological signature, called a persistence diagram, for each orbit from sampled data points [90]. Since diffeomorphic manifolds must be homeomorphic, their persistence diagrams must be similar. The stability of these diagrams ensures that the distance between them (e.g., the bottleneck distance) provides a robust measure of topological dissimilarity [91]. If this distance is large, we can reject the hypothesis that the orbits are diffeomorphic. TDA has been actively utilized in neuroscience to reveal topological structures of neural manifolds [92, 93, 94], and methods like topological RSA (tRSA) provide a framework for such comparisons [95].
- **Evaluating geometric similarity.** While TDA confirms the underlying topology, a diffeomorphism preserves a richer geometric structure. A standard method in neuroscience for evaluating geometric similarity is Representational Similarity Analysis (RSA) [96], but conventional RSA typically requires a known one-to-one correspondence between the points being compared (e.g., neural representations of a fixed set of stimuli). For comparing orbit structures across subjects, where such a strict correspondence is not guaranteed, the Gromov-Wasserstein Optimal Transport (GWOT) framework is a more powerful and flexible tool [97]. This is because GWOT finds the optimal, structure-preserving mapping between the orbits without presupposing point-wise correspondence. A low GWOT cost therefore indicates a strong structural correspondence between the two orbits, serving as a powerful proxy for the geometric similarity implied by a diffeomorphism. GWOT has recently been used to compare psychological and neural representation structures across individuals [24, 98, 99, 100].

Stage 2: Exploring the structure of qualia signatures

In contrast to the rigid orbits, the geometry of the quotient space, the structure of qualia signatures, is plastic and learned, with no guaranteed similarity across individuals (Proposi-

tion 6). Therefore, the goal of this stage is not to verify a strict equivalence, but rather to empirically assess the degree of geometric similarity between the subjects’ learned structures of qualia signatures.

Given this focus, methods that are sensitive to metric structures are well-suited. Standard Representational Similarity Analysis (RSA) [96] and other neural alignment techniques [101] can be used to evaluate geometric similarity between the distance matrices derived from the quotient spaces. Furthermore, Gromov-Wasserstein Optimal Transport (GWOT) is also particularly useful here, because it can compare structured metric spaces and distinguish whether two representational structures match at the fine-item level or a coarser categorical level by finding an optimal mapping between them [102, 100].

Transforming structural comparison into a nuanced diagnostic tool

This two-stage protocol transforms structural comparison into a nuanced diagnostic tool. We will now have a deeper understanding of which aspects of qualia structures match across subjects and which do not, according to the following possible scenarios.

- A low cost in Stage 1 (orbit alignment) is theoretically expected, which validates the equivariant neural representation assumption.
- A high cost in Stage 1 indicates a violation of the equivariant neural representation assumption, meaning one or both subjects may have failed to learn a proper equivariant representation for the symmetry group G .
- A high cost in Stage 2 (quotient alignment) confirms the idiosyncratic nature of the learned representational structures, which is the expected baseline.
- A low cost in both Stage 1 and Stage 2 corresponds to a close resemblance of qualia structures across individuals, suggesting that they have similar subjective experiences. In particular, a low cost in Stage 2, which is not theoretically guaranteed, would suggest a convergence of conceptual structures due to factors like shared learning objectives, for example, based on languages (see Discussion).

Approximate and emergent equivariance

Although the propositions derived from principal bundle geometry may appear self-evident in a purely mathematical context, they make powerful, empirically testable claims about the structure of any system that learns equivariant representations. This formulation provides more than just a descriptive theory; it offers a concrete research program for discovering nontrivial properties and achieving a deeper understanding of the qualia structure of a given neural system. For instance, as we proposed in the previous section, assessing individual differences in terms of orbits and quotient spaces is one such experimental approach to pursue in the future. The core of this program involves systematically applying transformations from a symmetry group G to sensory inputs and measuring the resulting changes in the

neural representation space. Through this process, one can empirically map the geometry of the orbits and the quotient space.

However, the theoretical framework, which assumes perfect G -equivariance, is an idealization. In reality, any neural system will at best exhibit approximate equivariance.

This reality shifts the focus toward a crucial set of empirical questions: What types of equivariance actually emerge in these systems, and how strongly are they expressed? Answering these questions requires robust methods for quantifying the deviation from perfect equivariance.

In artificial networks, several metrics have been developed for this purpose. These include the G -empirical equivariance deviation (G-EED) [103], which directly quantifies the extent to which a model’s output deviates from perfect equivariance, and the Local Equivariance Error (LEE) [104], which is derived from the Lie derivative and connects local deviations to global equivariance properties (see [105] for a unified perspective). Beyond simply measuring the degree of equivariance for known symmetries, such as rotation or translation, active research programs are also working to discover hidden or unknown symmetries directly from data, independent of prior knowledge [106, 107, 108].

Extending analogous measurements to the brain is more challenging because its input-output mapping can only be sampled indirectly. Nevertheless, a recent study introduced an in-vivo framework that directly estimates local equivariance from large-scale neuronal recordings, revealing that cortical populations can interpolate between finely tuned and transformation-invariant codes [75]. This approach provides a concrete route to quantifying equivariance in biological circuits and establishes a natural benchmark for comparing the operational symmetries of brains with those of artificial systems.

Compositionality and hierarchical structures of qualia

While our framework has so far focused on the geometric structure induced by a single symmetry group, G , conscious experience is often compositional, involving multiple interacting symmetries. How these different symmetries combine to create unified, structured experiences is a key question. Here, we extend our framework to model this compositionality using two fundamental algebraic structures: the direct product, which describes independent, disentangled qualia attributes, and the semidirect product, which describes hierarchically compositional, entangled qualia attributes.

For perceptually independent features, the total symmetry can be modeled as a direct product of groups ($G_{\text{total}} = G_1 \times G_2$). This structure corresponds to a direct product of principal bundles, a concept central to disentangled representation learning in AI [109, 110, 111, 112, 113]. In such a system, where the group $G_1 \times G_2$ acts component-wise on the product space $Y_1 \times Y_2$, the quotient space of signatures naturally factorizes into a product of the individual quotient spaces (Figure 5A):

$$(Y_1 \times Y_2)/(G_1 \times G_2) \cong (Y_1/G_1) \times (Y_2/G_2) =: Q_1 \times Q_2 \quad (3)$$

This factorization provides a formal model for how a neural system can, in principle, separate independent qualia attributes. At the highest level, the total experiential space Y might decompose into independent modalities such as vision and audition ($Y \cong Y_v \times Y_a$). Even within a single modality, a network could learn perfectly disentangled representations for an object’s pose and color ($G_v \cong G_{\text{pose}} \times G_{\text{color}}$), where the qualia attributes of pose and color could be experienced independently and commutatively (Figure 5B). However, achieving such ideal factorization in practice remains a significant challenge [30, 114].

While the direct product is a useful idealization, a unified percept often requires tighter, hierarchical integration of features. This feature entanglement is modeled by the semidirect product, $G_{\text{total}} = G_1 \rtimes_{\phi} G_2$, defined by the group operation $(g_1, g_2) \cdot (g'_1, g'_2) = (g_1 \cdot \phi_{g_2}(g'_1), g_2 \cdot g'_2)$, where $\phi : G_2 \rightarrow \text{Aut}(G_1)$ is a homomorphism specifying how G_2 elements modify those of G_1 . This structure is fundamentally non-commutative, reflecting hierarchical dependence: G_2 actions alter G_1 ’s effective transformations, but not vice versa. Geometrically, it induces a composite bundle, a hierarchy of bundles, with a two-stage projection [115]:

$$Y \rightarrow Q_1 \rightarrow Q_{12}. \quad (4)$$

The total space Y first forms a bundle under the action of the group G_1 , which partitions Y into orbits and yields the quotient space Q_1 . Then, this quotient space Q_1 itself serves as a new space that is acted upon by the second group, G_2 . This second action organizes Q_1 into another bundle structure, yielding a final quotient space, Q_{12} . A classic example is the combination of object translation ($G_1 = \mathbb{R}^2$) and rotation ($G_2 = SO(2)$) (Figure 5C), forming the special Euclidean group $SE(2)$. **Perceptually, rotating an object changes its effective position non-commutatively; that is, translating then rotating differs from rotating then translating.** This mirrors how visual stability (e.g., object constancy under motion) feels “entangled,” with position attributes depending on orientation context.

In summary, these two algebraic structures provide a richer language for modeling different modes of qualia composition. The direct product models the independence of qualia attributes through a commutative structure, corresponding geometrically to a simple, factorizable bundle ideal for disentangled experiences. In contrast, the semidirect product models the dependence and entanglement of qualia attributes, leading to a hierarchical, composite bundle geometry that explains how different attributes can be woven together into a single, coherent, and unified conscious experience.

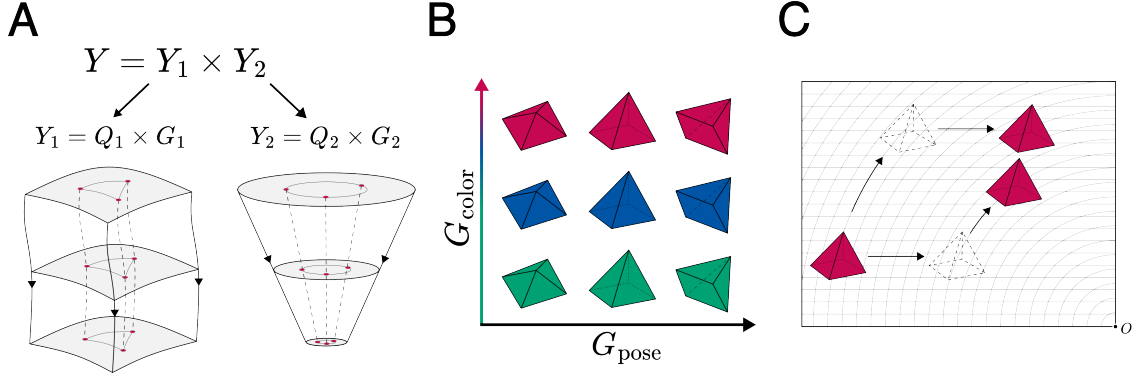


Figure 5: Compositionality and disentanglement of qualia. (A) The decomposition of Y under disentangled symmetry group $G = G_1 \times G_2$. Each factor space Y_1 and Y_2 is structured as a bundle over its respective quotient space.

(B) A conceptual example of a disentangled representation where color and pose vary independently along orthogonal axes. This reflects a direct product symmetry group $G_{\text{color}} \times G_{\text{pose}}$, where each axis corresponds to transformations under a distinct subgroup action. (C) Illustration of non-commutativity in the semidirect product $SE(2) = \mathbb{R}^2 \rtimes SO(2)$. Rotating the red tetrahedron (initially at lower left) around the frame’s lower-right corner (indicated by O) and then translating it right yields a different result than translating first and then rotating.

Conclusion and Discussion

In this Perspective, we have proposed that the relational structure of qualia can be rigorously described by the mathematical framework of principal bundle geometry.

This geometric framework provides a unified, three-level characterization of subjective experience: the qualia modality is determined by the algebraic structure of the group G itself; the qualia signature is represented by a point in the bundle’s quotient space Q ; and the qualia attribute is represented by a point on an orbit $\mathcal{O}$ induced by the group G .

While this framework provides a novel theoretical lens through which to view qualia, it also opens up numerous empirical avenues for future research.

The immediate priority of our framework is to undergo empirical validation through the proposed two-stage testing protocol. Applying methods like Topological Data Analysis

or Gromov-Wasserstein Optimal Transport on neural and behavioral data will be crucial for verifying the predicted universal topological structure of qualia attributes (orbits). The more exploratory second stage will assess the similarity in the geometric structure of qualia signatures (quotient spaces) across individuals. While our theory posits that the geometry in quotient space is plastic with no guaranteed inter-subject similarity, a growing body of evidence suggests that the neural representations of object categories can be remarkably consistent across individuals [116, 117]. We hypothesize that this alignment of neural representations is driven by shared pressures among humans, the most significant of which is language. This claim resonates with modern, nuanced interpretations of the Sapir-Whorf hypothesis, also known as linguistic relativity, which suggests that the languages we speak shape how we perceive, remember, and categorize the world [118]. By providing a public framework for negotiating and stabilizing shared meanings, language may therefore act as a powerful scaffold that constrains the representation learning process [119, 120]. Indeed, the recent success of large-scale vision-language models suggests that grounding visual representations in a shared linguistic space promotes the emergence of a structured semantic geometry [121]. This, in turn, may compel different individuals to develop similar relational geometries for their qualia signatures, thus aligning their otherwise idiosyncratic quotient spaces.

Although few theories of consciousness attempt to provide a principled account of the qualitative differences between our various experiences [16, 17], making our geometric framework based on symmetry a unique contribution, Integrated Information Theory (IIT) represents a notable and pioneering research program that connects the structure of a physical substrate to the quality of experience [122, 123, 124]. A critical difference between our framework and IIT is the distinction between feed-forward information processing, which connects external input space to neural representation space, and intrinsic recurrent information processing. IIT posits that consciousness is identical to a maximally irreducible cause-effect structure that arises from the intrinsic causal interactions of recurrent networks. In contrast, our theory explains the relational structure of qualia as the geometric structure induced by G -equivariant feed-forward functions that learn and preserve extrinsic symmetries from the input space. This divergence is clear in their explanations for the experience of space and time. IIT proposes that the extended feeling of space and the directed flow of time arise from the different types of recurrent interactions in different neural architectures: a non-directed grid for space versus a directed grid for time [125, 126]. On the other hand, our framework would attribute the structure of spatial experience to the geometric structure inherited from external symmetry groups, such as translation and rotation, through feed-forward processing. While both theories ground qualia in structure, the relationship between IIT’s intrinsic causal structures and our framework’s group-induced geometric structures remains an open question.

In conclusion, the application of principal bundle geometry to the study of qualia offers a rich, mathematically grounded, and empirically testable approach. By focusing on the role of symmetry in shaping neural representations, it provides a unifying perspective that connects our understanding of perception from the level of group theory, through neural

population activity, to the phenomenology of subjective experience. We believe that this geometric language holds significant promise for demystifying the structural aspects of qualia and for guiding future research into the neural basis of consciousness.

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Supplementary Information for “Principal bundle geometry of qualia: Understanding the quality of consciousness from symmetry”

Contents

- Definition of a principal bundle
 - Proofs of propositions
-

Definition of a principal bundle

In the main text, we propose that the geometry of qualia is characterized by a principal bundle. Here, we provide a formal definition of this mathematical structure.

A principal bundle consists of a total space Y , a base space Q , a Lie group G , and a continuous surjective map $\pi : Y \rightarrow Q$ that satisfies the following conditions:

1. **Group action:** The group G acts on the total space Y from the right. This action is smooth and free, meaning that for any point $y \in Y$, if $y \cdot g = y$ for some $g \in G$, then g must be the identity element of the group.
2. **Projection and orbits:** The projection map π is constant on the orbits of the group action. This means that for any point $y \in Y$ and any group element $g \in G$, $\pi(y \cdot g) = \pi(y)$. Consequently, the base space Q is the quotient space of Y by the group action, written as $Q = Y/G$. The pre-image $\pi^{-1}(q)$ for any point $q \in Q$ is an orbit of the group action and is called a fiber.
3. **Local triviality:** The bundle is locally a product space. For any point $q \in Q$, there exists an open neighborhood U of q in the base space such that the pre-image $\pi^{-1}(U)$ is homeomorphic to the Cartesian product $U \times G$. This homeomorphism, $\phi : \pi^{-1}(U) \rightarrow U \times G$, is required to be equivariant in the sense that $\phi(y \cdot g) = (\pi(y), \phi_G(y) \cdot g)$, where $\phi_G(y)$ is the group component of $\phi(y)$. This condition ensures that the structure of the fibers is locally consistent and identical to the structure of the group G itself.

In the context of our theory, the neural representation space Y is the total space, the symmetry group G defines the qualia modality, the orbits (fibers) represent qualia attributes, and the quotient space (base space) Q represents qualia signatures.

Proofs of propositions

Rigidity of orbits

Proposition 1 (Orbit-group diffeomorphism). Let $f : X \rightarrow Y$ be a smooth, proper, G -equivariant map between smooth G -spaces X and Y . Assume the action on X is free and proper. Then for any $x_0 \in X$, if the action of G on Y is free at $y_0 = f(x_0)$, the orbit $\mathcal{O}_Y(y_0)$ is a properly embedded smooth submanifold of Y that is diffeomorphic to G .

Proof. Let $y_0 \in \text{Im}(f)$, and choose a point $x_0 \in X$ such that $f(x_0) = y_0$. The proof demonstrates that the orbit map from G to the orbit of y_0 is a proper embedding, which establishes the result.

The orbit map is a smooth injective immersion. The orbit of y_0 is the image of the orbit map $\phi_{y_0} : G \rightarrow Y$, defined by $\phi_{y_0}(g) = \rho_Y(g)y_0$. The map ϕ_{y_0} is smooth because it is the composition of the smooth map $g \mapsto (g, y_0)$ and the smooth action map $G \times Y \rightarrow Y$.

We are given that the action on Y is free at y_0 , meaning the isotropy group $G_{y_0} = \{g \in G \mid \rho_Y(g)y_0 = y_0\}$ is trivial. This implies ϕ_{y_0} is injective.

By Proposition 7.26 [36], every orbit map of a smooth Lie group action has constant rank. Since ϕ_{y_0} is an injective smooth map of constant rank, the Global Rank Theorem (Theorem 4.14(b) [36]) implies it is a smooth immersion.

The orbit map is a proper map. A map is proper if the preimage of every compact set is compact. The equivariance of f is used here to relate the actions on X and Y .

- The G -equivariance of f establishes the identity $\phi_{y_0} = f \circ \phi_{x_0}$, where $\phi_{x_0}(g) = \rho_X(g)x_0$.
- We are given that the action of G on X is proper. By Proposition 21.7 [36], this implies that the orbit map $\phi_{x_0} : G \rightarrow X$ is a proper map.
- We are also given that the map $f : X \rightarrow Y$ is a proper map.
- The composition of two proper maps is proper. Therefore, the map $\phi_{y_0} = f \circ \phi_{x_0}$ is a proper map from G to Y .

Conclusion. We have shown that $\phi_{y_0} : G \rightarrow Y$ is a proper, injective, smooth immersion.

- By Proposition 4.22(b) [36], an injective proper immersion is a smooth embedding.
- The image of a smooth embedding is an embedded submanifold (Proposition 5.2 [36]). As the image of ϕ_{y_0} is the orbit $\mathcal{O}_Y(y_0)$, the orbit is an embedded submanifold of Y .
- Because ϕ_{y_0} is a proper map and Y is a locally compact Hausdorff space, its image $\mathcal{O}_Y(y_0)$ is a closed subset of Y (Theorem A.57 [36]). An embedded submanifold that is also a closed subset is a properly embedded submanifold (Proposition 5.5 [36]).

- Finally, since ϕ_{y_0} is a smooth embedding, it is a diffeomorphism from its domain G onto its image $\mathcal{O}_Y(y_0)$.

Thus, the orbit $\mathcal{O}_Y(y_0)$ is a properly embedded smooth submanifold of Y that is diffeomorphic to the Lie group G . $\square$

Proposition 2 (Inter-subject orbit diffeomorphism). Let Person A have an equivariant encoder $f_A : X \rightarrow Y_A$ (with representation ρ_A) and Person B have an equivariant encoder $f_B : X \rightarrow Y_B$ (with representation ρ_B), both with respect to the same Lie group G . Let $x_0 \in X$ be a point in the stimulus space, and let $y_A = f_A(x_0) \in Y_A$ and $y_B = f_B(x_0) \in Y_B$ be the corresponding neural representations. If the G -actions on Y_A and Y_B are free at y_A and y_B respectively, then the orbit $\mathcal{O}_{Y_A}(y_A)$ is diffeomorphic to the orbit $\mathcal{O}_{Y_B}(y_B)$.

Proof. The objective is to demonstrate that the orbits $\mathcal{O}_{Y_A}(y_A)$ and $\mathcal{O}_{Y_B}(y_B)$ are diffeomorphic. This proof hinges on the principle that if two spaces are each diffeomorphic to a third common space, then they are diffeomorphic to each other. In this context, the common third space is the Lie group G .

1. **Application of Proposition 1:** Proposition 1 states that for a G -equivariant system where the Lie group G acts smoothly, freely, and properly on a smooth manifold Y , any orbit $\mathcal{O}_Y(y_0)$ (for $y_0 \in Y$ in the image of the encoder) is a smooth submanifold of Y that is diffeomorphic to G . The orbit map $\phi_Y : G \rightarrow \mathcal{O}_Y(y_0)$, defined by $\phi_Y(g) = \rho(g)y_0$, is the diffeomorphism that establishes this relationship.
 - For Person A's system: Given the anchor point $y_A = f_A(x_0)$, and assuming the G -action on Y_A is free at y_A , Proposition 1 ensures the existence of a diffeomorphism $\phi_A : G \rightarrow \mathcal{O}_{Y_A}(y_A)$.
 - For Person B's system: Similarly, with the anchor point $y_B = f_B(x_0)$, and given that the G -action on Y_B is free at y_B , Proposition 1 implies the existence of a diffeomorphism $\phi_B : G \rightarrow \mathcal{O}_{Y_B}(y_B)$.
2. **Construction of the inter-subject diffeomorphism:** Since both $\mathcal{O}_{Y_A}(y_A)$ and $\mathcal{O}_{Y_B}(y_B)$ are diffeomorphic to the same Lie group G , it follows that $\mathcal{O}_{Y_A}(y_A)$ is diffeomorphic to $\mathcal{O}_{Y_B}(y_B)$. An explicit diffeomorphism $\Psi : \mathcal{O}_{Y_A}(y_A) \rightarrow \mathcal{O}_{Y_B}(y_B)$ can be constructed by composing ϕ_B with the inverse of ϕ_A , i.e., $\Psi = \phi_B \circ \phi_A^{-1}$. A diffeomorphism is defined as a bijective smooth map with a smooth inverse. The inverse of a diffeomorphism is itself a diffeomorphism, and the composition of two diffeomorphisms is also a diffeomorphism. Therefore, Ψ is indeed a diffeomorphism, completing the proof. $\square$

Proposition 3 (Orbit-group isometry). Suppose the neural representation space Y is endowed with a Riemannian metric g_Y (for example, the Euclidean metric). If the representation $\rho : G \rightarrow O(d)$ is an orthogonal representation, then for any $y_0 \in Y$ in the image of f , the orbit $\mathcal{O}_Y(y_0)$, endowed with the induced Riemannian metric, is isometric to the Lie

group G , endowed with the pullback Riemannian metric $\phi_Y^* g_Y$. Furthermore, this pullback metric $\phi_Y^* g_Y$ on G is a left-invariant Riemannian metric.

Proof. Let g_Y be the Riemannian metric on the ambient space Y . The orbit $\mathcal{O}_Y(y_0)$ inherits an induced Riemannian metric from Y , which we also denote by g_Y . Let g_G denote the pullback metric on the Lie group G , defined by $g_G = \phi_{y_0}^* g_Y$, where $\phi_{y_0} : G \rightarrow \mathcal{O}_Y(y_0)$ is the orbit map defined by $\phi_{y_0}(g) = \rho(g)y_0$.

1. The orbit-group map ϕ_{y_0} is an isometry. An isometry is a diffeomorphism that preserves the Riemannian metric. From Proposition 1, we know that the orbit map $\phi_{y_0} : G \rightarrow \mathcal{O}_Y(y_0)$ is a diffeomorphism. We must show it preserves the metric. For any tangent vectors $X, Z \in T_g G$ at a point $g \in G$, the definition of the pullback metric is:

$$(g_G)_g(X, Z) := (g_Y)_{\phi_{y_0}(g)}(d(\phi_{y_0})_g(X), d(\phi_{y_0})_g(Z)) \quad (5)$$

This equation states that the inner product of two vectors in G (with metric g_G) is defined to be the inner product of their images under the differential $d\phi_{y_0}$ in the orbit (with metric g_Y). By its very construction, the map $\phi_{y_0} : (G, g_G) \rightarrow (\mathcal{O}_Y(y_0), g_Y)$ is an isometry.

2. The pullback metric g_G is left-invariant. This is the main result, and it relies critically on the representation ρ being orthogonal. A metric g_G is left-invariant if, for every $h \in G$, the left-translation map $L_h : G \rightarrow G$ is an isometry, i.e., $L_h^* g_G = g_G$. We evaluate the pullback of g_G by L_h at an arbitrary point $g \in G$ on arbitrary tangent vectors $X, Z \in T_g G$:

$$(L_h^* g_G)_g(X, Z) = (g_G)_{L_h(g)}(d(L_h)_g(X), d(L_h)_g(Z)) = (g_G)_{hg}(d(L_h)_g(X), d(L_h)_g(Z)) \quad (6)$$

Now, we substitute the definition of $g_G = \phi_{y_0}^* g_Y$:

$$= (g_Y)_{\phi_{y_0}(hg)}(d(\phi_{y_0})_{hg}(d(L_h)_g(X)), d(\phi_{y_0})_{hg}(d(L_h)_g(Z))) \quad (7)$$

Using the chain rule, $d(\phi_{y_0})_{hg} \circ d(L_h)_g = d(\phi_{y_0} \circ L_h)_g$. From the group action, we have the relation $\phi_{y_0}(hg) = \rho(h)(\phi_{y_0}(g))$, which means $\phi_{y_0} \circ L_h = \rho(h) \circ \phi_{y_0}$. Taking the differential at g :

$$d(\phi_{y_0} \circ L_h)_g = d(\rho(h) \circ \phi_{y_0})_g = d(\rho(h))_{\phi_{y_0}(g)} \circ d(\phi_{y_0})_g \quad (8)$$

Since $\rho(h)$ is a linear map, its differential is just itself. So, $d(\phi_{y_0} \circ L_h)_g = \rho(h) \circ d(\phi_{y_0})_g$. Substituting this back into our main expression gives:

$$(L_h^* g_G)_g(X, Z) = (g_Y)_{\rho(h)(\phi_{y_0}(g))}(\rho(h)(d(\phi_{y_0})_g(X)), \rho(h)(d(\phi_{y_0})_g(Z))) \quad (9)$$

This is where we use the crucial assumption that ρ is an **orthogonal representation**. By definition, this means that for any $h \in G$, the linear map $\rho(h)$ is an orthogonal transformation and preserves the inner product g_Y . Therefore:

$$(g_Y)_{\rho(h)(\phi_{y_0}(g))}(\rho(h)(v_1), \rho(h)(v_2)) = (g_Y)_{\phi_{y_0}(g)}(v_1, v_2) \quad (10)$$

Applying this property, our expression simplifies to:

$$(L_h^* g_G)_g(X, Z) = (g_Y)_{\phi_{y_0}(g)}(d(\phi_{y_0})_g(X), d(\phi_{y_0})_g(Z)) \quad (11)$$

By the definition of the pullback metric, this is exactly $(g_G)_g(X, Z)$. Since this holds for all g, X, Z , we have shown $L_h^* g_G = g_G$, which proves that the pullback metric is left-invariant. $\square$

Proposition 4 (Uniform metric distortion of inter-subject orbit maps). Let $\mathcal{O}_{Y_A}(y_A)$ and $\mathcal{O}_{Y_B}(y_B)$ be two orbits generated by orthogonal representations. The entire geometric relationship between them is determined by a single, constant, self-adjoint linear operator $L : \mathfrak{g} \rightarrow \mathfrak{g}$ on the Lie algebra $\mathfrak{g}$, which relates the two pullback metrics at the identity: $\langle X, Z \rangle_B = \langle X, L(Z) \rangle_A$. In the special case where this operator is a scalar multiple of the identity, $L = c \cdot \text{Id}$, the map Ψ is a homothety.

Proof. The proof proceeds by first establishing that the geometry of each orbit is determined by a single inner product on the Lie algebra $\mathfrak{g}$. Then, it shows that the relationship between these two inner products is a constant linear operator, which in turn dictates the nature of the diffeomorphism Ψ between the orbits.

1. Left-invariant metrics on G . From Proposition 3, we know that because the representations $\rho_A : G \rightarrow O(d_A)$ and $\rho_B : G \rightarrow O(d_B)$ are orthogonal, the pullback metrics on the Lie group G , defined as $g_A = \phi_A^* g_{Y_A}$ and $g_B = \phi_B^* g_{Y_B}$, are both left-invariant Riemannian metrics. A left-invariant metric is uniquely determined by its value at the identity element $e \in G$. This value is an inner product on the Lie algebra $\mathfrak{g} = T_e G$. We denote these two inner products as $\langle \cdot, \cdot \rangle_A = (g_A)_e$ and $\langle \cdot, \cdot \rangle_B = (g_B)_e$.

2. The linear operator relating the geometries. We now have two distinct inner products, $\langle \cdot, \cdot \rangle_A$ and $\langle \cdot, \cdot \rangle_B$, defined on the same finite-dimensional vector space $\mathfrak{g}$. By a standard result from linear algebra, for any two such inner products, there exists a unique linear operator $L : \mathfrak{g} \rightarrow \mathfrak{g}$ which is self-adjoint with respect to $\langle \cdot, \cdot \rangle_A$ and relates the two inner products as follows for all $X, Z \in \mathfrak{g}$:

$$\langle X, Z \rangle_B = \langle X, L(Z) \rangle_A \quad (12)$$

This single, constant linear operator L completely encapsulates the difference between the two metric geometries at the infinitesimal level of the Lie algebra.

3. From the Lie algebra to the orbits. The diffeomorphism between the two orbits is given by $\Psi = \phi_B \circ \phi_A^{-1}$. The geometric distortion induced by this map is characterized by the pullback metric $\Psi^* g_{Y_B}$. We can relate this back to the metrics on the group G :

$$\Psi^* g_{Y_B} = (\phi_B \circ \phi_A^{-1})^* g_{Y_B} = (\phi_A^{-1})^* (\phi_B^* g_{Y_B}) = (\phi_A^{-1})^* g_B \quad (13)$$

Because both g_A and g_B are left-invariant, the relationship between them is uniform across the entire group. The linear relationship L at the identity extends globally. That is, at any point $g \in G$ and for any left-invariant vector fields X, Y , we have $(g_B)_g(X, Y) = (g_A)_g(X, L(Y))$. This means the distortion is not local but uniform.

4. The special case of homothety. We now consider the special case where the operator L is a scalar multiple of the identity, i.e., $L = c \cdot \text{Id}$ for some constant $c > 0$. In this case, the relationship between the inner products on the Lie algebra simplifies to:

$$\langle X, Z \rangle_B = \langle X, c \cdot Z \rangle_A = c \langle X, Z \rangle_A \quad (14)$$

By left-invariance, this constant proportionality extends to the entire group, so $g_B = c \cdot g_A$. Substituting this into our pullback expression from paragraph 3:

$$\Psi^* g_{Y_B} = (\phi_A^{-1})^* g_B = (\phi_A^{-1})^* (c \cdot g_A) = c \cdot (\phi_A^{-1})^* g_A \quad (15)$$

From Proposition 3, the map $\phi_A : (G, g_A) \rightarrow (\mathcal{O}_A, g_{Y_A})$ is an isometry. Therefore, its inverse ϕ_A^{-1} is also an isometry, and the pullback of g_A by ϕ_A^{-1} is the original induced metric on the orbit, g_{Y_A} .

$$(\phi_A^{-1})^* g_A = g_{Y_A} \quad (16)$$

This yields the final result for the special case:

$$\Psi^* g_{Y_B} = c \cdot g_{Y_A} \quad (17)$$

This is the definition of a homothety. $\square$

Plasticity of quotient space

Proposition 5 (Quotient plasticity). Let S be a smooth global section of Y corresponding to a G -equivariant encoder f . The intrinsic geometry of S is not fixed by the G -equivariance constraint alone. By composing the encoder f with a valid G -equivariant linear map, the geometry of the resulting section $\tilde{S}$ can be arbitrarily altered.

Proof. The proposition asserts that the intrinsic metric geometry of a section (representing qualia signatures) is a plastic and learnable property of the neural representation. We prove this by demonstrating that a valid G -equivariant linear map can arbitrarily alter the geometry of a given section, a mechanism rooted in the nature of group representations.

1. **Initial setup:** Let $f : X \rightarrow Y$ be an initial G -equivariant encoder, and let $s : Q_Y \rightarrow Y$ be a corresponding smooth global section. Its image, $S = s(Q_Y)$, is an embedded submanifold of Y that inherits its intrinsic geometry (distances and angles) from the Euclidean metric of the ambient space Y .

2. The role of G -equivariant linear maps:

- Consider a G -equivariant linear map $L : Y \rightarrow Y$. Such a map commutes with the group action, meaning $L \circ \rho(g) = \rho(g) \circ L$ for all $g \in G$, where ρ is the representation of G on Y .
- If Person A modifies their encoder by composing it with such a map, say $L_A : Y_A \rightarrow Y_A$, they form a new valid encoder $\tilde{f}_A = L_A \circ f_A$. This transformation maps the original section S_A to a new submanifold $\tilde{S}_A = L_A(S_A)$, which serves as a valid section for the new encoder.

3. **Arbitrary alteration of geometry:** We now show that the intrinsic geometry of $\tilde{S}$ is generally different from that of S . Let y_1, y_2 be two distinct points within the original section S . The squared distance between them, which defines the metric on S , is $\|y_1 - y_2\|_2^2$.

These points are mapped to $\tilde{y}_1 = L(y_1)$ and $\tilde{y}_2 = L(y_2)$ in the new section $\tilde{S}$. The squared distance between these new points is:

$$\|\tilde{y}_1 - \tilde{y}_2\|_2^2 = \|L(y_1) - L(y_2)\|_2^2 = \|L(y_1 - y_2)\|_2^2.$$

According to Schur's Lemma (see [127] for example), if the representation ρ is irreducible, any such commuting map L must be a scalar multiple of the identity, $L = \lambda I$, which would only scale the geometry uniformly. However, representations in neural systems are generally reducible. This means the representation space Y can be decomposed into a direct sum of irreducible subspaces, $Y = V_1 \oplus V_2 \oplus \dots \oplus V_k$. The G -equivariant map L can then be a block-diagonal matrix, applying a different scaling factor λ_i to each subspace V_i .

Unless the vector $(y_1 - y_2)$ lies entirely within a single irreducible subspace where $|\lambda_i| = 1$, the map L will change its length. Since vectors connecting points within a section are not generally constrained in this way, one can choose the scaling factors (λ_i) to arbitrarily expand, contract, or warp the distances within the section.

Therefore, the intrinsic geometry of a section is not fixed by G -equivariance alone, but can be arbitrarily altered by composing the encoder with a G -equivariant linear map, due to the reducibility of neural representations. □

Proposition 6 (Inter-subject quotient plasticity). Let Person A and Person B have encoders f_A and f_B that are both equivariant with respect to the same group G . Let S_A and S_B be any corresponding sections of Y_A and Y_B , respectively. There is no guaranteed geometric relationship (e.g., isometry, similarity, or even diffeomorphism) between the submanifolds S_A and S_B with their inherited Riemannian metrics.

Proof. The proposition states that no specific geometric relationship is guaranteed to hold between corresponding sections S_A and S_B from two different subjects. We prove this by demonstrating that even if a geometric relationship did exist for a given pair of encoders, it can be broken by a valid, equivariance-preserving transformation on one of the subjects.

1. Initial setup:

- Let Person A and Person B have G -equivariant encoders, $f_A : X \rightarrow Y_A$ and $f_B : X \rightarrow Y_B$, respectively. Y_A and Y_B are neural representation spaces (Euclidean spaces $\mathbb{R}^{d_A}$ and $\mathbb{R}^{d_B}$), and the encoders induce representations $\rho_A : G \rightarrow GL(Y_A)$ and $\rho_B : G \rightarrow GL(Y_B)$.
- Let $s_A : Q_A \rightarrow Y_A$ and $s_B : Q_B \rightarrow Y_B$ be smooth global sections for each subject's representation. Their images, $S_A = s_A(Q_A)$ and $S_B = s_B(Q_B)$, are

embedded submanifolds of Y_A and Y_B , respectively. These submanifolds inherit a Riemannian metric from the Euclidean metric of their ambient spaces.

2. **Assume a relationship exists:** For the sake of argument, suppose that for a specific choice of encoders and sections, the resulting manifolds S_A and S_B happen to be isometric. This means there exists a map $\Phi : S_A \rightarrow S_B$ such that for any two points $y_1, y_2 \in S_A$, their squared distance is preserved: $\|\Phi(y_1) - \Phi(y_2)\|^2 = \|y_1 - y_2\|^2$.

3. **Apply a plasticity transformation:**

- According to Proposition 5, Person A can modify their encoder by composing it with a valid G -equivariant linear map $L_A : Y_A \rightarrow Y_A$ to form a new valid encoder, $\tilde{f}_A = L_A \circ f_A$.
- This transformation maps the original section S_A to a new submanifold $\tilde{S}_A = L_A(S_A)$. The new submanifold $\tilde{S}_A$ is a valid section for the new representation space defined by the encoder $\tilde{f}_A$.

4. **The relationship is broken:**

- Now, consider the geometry of the new section $\tilde{S}_A$. Let y_1, y_2 be two points in the original section S_A . Their images in the new section are $\tilde{y}_1 = L_A(y_1)$ and $\tilde{y}_2 = L_A(y_2)$. The squared distance between these new points is: $\|\tilde{y}_1 - \tilde{y}_2\|_2^2 = \|L_A(y_1) - L_A(y_2)\|_2^2 = \|L_A(y_1 - y_2)\|_2^2$.
- As established in the proof of Proposition 5, if the representation ρ_A is reducible, a G -equivariant linear map L_A can be chosen such that it is not a simple scalar multiple of the identity or a rotation. Specifically, L_A can scale different irreducible subspaces of Y_A by different factors.
- In such a case, there is no guarantee that $\|L_A(y_1 - y_2)\|^2 = \|y_1 - y_2\|^2$. Since the internal distances within the section S_A have been altered by the map L_A , the new section $\tilde{S}_A$ is generally not isometric to the original section S_A . Consequently, it cannot be isometric to S_B either (unless by sheer coincidence, which would not constitute a “guaranteed relationship”).

Since for any given pair of encoders, one can find a valid subsequent transformation for one subject that changes the intrinsic geometry of their section, no specific geometric relationship (isometry, similarity, or even diffeomorphism) between the sections of two different subjects is guaranteed by G -equivariance alone. The geometry of the section is a plastic, learnable feature of the representation. $\square$

Proposition 7 (Intra-subject quotient diffeomorphism). Let $s_1 : Q_Y \rightarrow Y$ be a smooth global section, and let $S_1 = s_1(Q_Y)$ be its image. For any group element $g \in G$, let S_2 be the image of the section $s_2 = \rho(g) \circ s_1$. The map $\Psi_g : S_1 \rightarrow S_2$ defined by restricting the group action, $\Psi_g(y) = \rho(g)y$, is a diffeomorphism.

Proof. To prove this, we will demonstrate that the map Ψ_g is a bijective smooth map with a smooth inverse, which are the defining characteristics of a diffeomorphism. This directly

follows from Ψ_g being a restriction of a G -action, where each $\rho(g)$ is an invertible linear transformation, and such transformations are known to be diffeomorphisms.

1. Preliminaries: smooth manifold structure of Q_Y , S_1 , and S_2 :

- The framework presupposes that Q_Y is a smooth manifold. The existence of a smooth global section $s_1 : Q_Y \rightarrow Y$ is consistent with the principal bundle being smoothly trivial.
- For $S_1 = s_1(Q_Y)$ to be an embedded submanifold of Y , it must be the image of a smooth embedding. A smooth global section s_1 is typically defined as a smooth map such that $\pi \circ s_1 = Id_{Q_Y}$. For s_1 to induce an embedded submanifold structure on its image, it must be a smooth injective immersion that maps Q_Y homeomorphically onto its image with the subspace topology. We assume s_1 satisfies these conditions, thus S_1 is a smooth manifold.
- The map $L_g : Y \rightarrow Y$ defined by $L_g(y) = \rho(g)y$ is smooth, as $\rho : G \times Y \rightarrow Y$ is a smooth group action (a fundamental property of Lie groups and their representations). The section s_2 is the composition of two smooth maps, L_g and s_1 . Therefore, $s_2 = L_g \circ s_1$ is also a smooth map. Since L_g is an invertible linear transformation (as $\rho(g) \in GL(Y)$), it is a diffeomorphism of Y onto itself. Consequently, s_2 is also a smooth embedding, making $S_2 = s_2(Q_Y)$ an embedded submanifold of Y , and thus a smooth manifold.

2. **Smoothness of Ψ_g :** The map $\Psi_g : S_1 \rightarrow Y$ is the restriction of the smooth map $L_g : Y \rightarrow Y$ to the smooth embedded submanifold S_1 . The restriction of a smooth map to a smooth submanifold is smooth (Theorem 5.53(a) in [36]). Therefore, $\Psi_g : S_1 \rightarrow Y$ is smooth. Since its image is precisely S_2 , Ψ_g can be considered as a smooth map from S_1 to S_2 .

3. Bijectivity of Ψ_g :

- **Injectivity:** Suppose $\Psi_g(y_1) = \Psi_g(y_2)$ for $y_1, y_2 \in S_1$. This means $\rho(g)y_1 = \rho(g)y_2$. As ρ is a linear representation of the Lie group G , each $\rho(g)$ is an invertible linear transformation (an element of $GL(Y)$). Thus, $\rho(g)$ has an inverse, which is $\rho(g^{-1})$. Applying $\rho(g^{-1})$ to both sides of the equation yields $\rho(g^{-1})\rho(g)y_1 = \rho(g^{-1})\rho(g)y_2$. By the group homomorphism property, $\rho(g^{-1})\rho(g) = \rho(g^{-1}g) = \rho(e)$, where e is the identity element of G . The identity element of a group representation maps to the identity linear transformation, i.e., $\rho(e) = Id_Y$. Therefore, $y_1 = y_2$. This proves Ψ_g is injective.
- **Surjectivity:** Let $z \in S_2$ be an arbitrary point. By the definition of S_2 , there exists a point $q \in Q_Y$ such that $z = s_2(q)$. Since $s_2 = \rho(g) \circ s_1$, we can write $z = \rho(g)(s_1(q))$. Let $y_1 = s_1(q)$. By definition of S_1 , $y_1 \in S_1$. Substituting this into the expression for Ψ_g , we find $\Psi_g(y_1) = \rho(g)y_1 = \rho(g)(s_1(q)) = z$. This proves Ψ_g is surjective.
- Since Ψ_g is both injective and surjective, it is a bijection.

4. Smoothness of the Inverse Map Ψ_g^{-1} :

- The inverse map $\Psi_g^{-1} : S_2 \rightarrow S_1$ is given by $\Psi_g^{-1}(z) = \rho(g^{-1})z$.
- Analogous to the argument for Ψ_g itself, this inverse map is also a restriction of a smooth map ($L_{g^{-1}}(z) = \rho(g^{-1})z$) to a smooth embedded submanifold (S_2). Therefore, Ψ_g^{-1} is also smooth.

Since $\Psi_g : S_1 \rightarrow S_2$ is a smooth bijective map with a smooth inverse, it satisfies the definition of a diffeomorphism. $\square$